

Protocol Title: A Systems Biology Study to Investigate Immune Correlates of the Response to Licensed Hepatitis B Vaccine (Engerix-B®) among Healthy Subjects 50 to 70 Years of Age

Protocol Number: DV2-HBV-E

No. of Centers: Up to 2 sites in the US

Sample Size: Up to 60 subjects

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Original Protocol: 17 February 2011

INVESTIGATOR SIGNATURE PAGE

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DECLARATION OF INVESTIGATOR

I confirm that I have read and understood this protocol, and agree to conduct the study as outlined in the protocol and other information supplied to me.

I agree to conduct the study in compliance with all local legal and regulatory requirements, and according to good clinical practice as described in the United States Code of Federal Regulations – Title 21, Parts 11, 50, 54, 56, 312 Subpart D; the International Conference on Harmonisation document “Guidance for Industry – E6 Good Clinical Practice: Consolidated Guidance”; and the Declaration of Helsinki.

Investigator Signature

Date

Investigator Name (Print)

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PROTOCOL SYNOPSIS

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Study Objectives Primary Objectives:

- Document the changes in cellular responses and gene expression patterns in serial blood samples collected from healthy older adults after vaccination with Engerix-B
- Determine which of these changes correlate with the development of protective levels of antibody (anti-HBsAg ≥ 10 milli-International units per milliliter [mIU/mL]) to hepatitis B virus (HBV)

Secondary Objective:

- Compare and contrast the cellular responses and gene expression patterns in healthy older adults after the first vs. second injection with Engerix-B

Study Design This is a single arm, open label study of up to 60 healthy adult subjects 50 to 70 years of age not previously vaccinated with a HBV vaccine. After providing informed consent and meeting eligibility criteria, subjects will receive a total of 3 injections of Engerix-B per the recommended adult schedule, administered at Weeks 0, 4, and 24 (Visits 1, 5 and 10). After each of the first 2 injections at Visits 1 and 5, subjects will be asked to return at specified times for the collection of blood samples for the investigational studies. After the third injection, samples will only be collected for the measurement of antibody response to HBV.

Study Population The population for this study is healthy HBV naïve males and females 50 to 70 years of age, inclusive.

Study Period Enrollment and signing of informed consent forms will occur between 1 and 21 days prior to the first injection of vaccine. Each subject will be given the 3-dose vaccination regimen (0, 4, and 24 weeks) as recommended by the manufacturer. Blood samples will be collected immediately prior to each of the first 2 injections of Engerix-B, and at 1, 3, 7 days after those injections. Each subject is expected to remain in the study through Week 28 (Visit 11), 4 weeks after the third Engerix-B injection.

Study Assessments The schedule of visits and study assessments are presented in detail in Appendix 1.

Blood samples will be drawn at Screening, Days 0, 1, 3, 7, 28, 29, 31, 35, 56, 168, and 196.

Immunology: Multiple approaches to humoral and cellular responses will be used including measurement of anti-HBs (anti-HBsAg [hepatitis B surface antigen]) levels, genome-wide transcriptional profile of messenger ribonucleic acid (mRNA), transcriptional response of peripheral blood mononuclear cells (PBMCs) to Toll-like receptor (TLR) ligands, phenotyping of B and T cells and dendritic cells, antigen restimulation and phenotyping of antigen-specific T cells, antigen restimulation and measurement of cytokine mRNA levels, and, possibly, analysis of genetic polymorphisms.

Study Treatment Engerix-B (20 mcg HBsAg, GlaxoSmithKline) is a FDA-approved HBV vaccine.

Dosage and Administration:

Subjects will receive 1.0 mL Engerix-B administered intramuscularly in the deltoid muscle at Week 0 (Visit 1), Week 4 (Visit 5), and Week 24 (Visit 10) as described in the manufacturer's Prescribing Information.

Eligibility Criteria Inclusion Criteria:

A subject must meet all of the following inclusion criteria to participate in the study:

1. be 50 to 70 years of age, inclusive;
2. be in good health in the opinion of the Investigator, based upon medical history, physical examination, and laboratory evaluation;
3. be able to comprehend and follow all required study procedures, and be available for all visits scheduled in the study;
4. be serum negative for hepatitis B surface antigen (HBsAg), HBsAg antibody (anti-HBs), antibody to hepatitis B core antigen (anti-HBcAg), hepatitis C virus (HCV), and human immunodeficiency virus (HIV);
5. if female of childbearing potential, agree to consistently use a highly effective method of birth control from the screening visit through Visit 11^a;
6. be able and willing to provide informed consent.

Exclusion Criteria:

A subject who meets any of the following criteria is not permitted to participate in the study:

1. if female of childbearing potential, is pregnant, breastfeeding, or planning a pregnancy;
2. has a known history of multiple sclerosis;
3. has previously received any hepatitis B vaccine (approved or investigational);
4. has a history of sensitivity to any component of Engerix-B;

^a Highly effective methods of birth control include: chemical contraceptive, intrauterine device (IUD), or condom plus spermicide. Abstinence will also be accepted but only with written confirmation from the subject.

5. has body-mass index (BMI) > 30, diabetes mellitus (Type I or II), or other chronic medical condition that in the opinion of the Investigator might interfere with the immune response to hepatitis B vaccination;
6. is unwilling or unable to comply with all the requirements of the protocol;
7. has current or suspected immunodeficiency disorder or treatment with immunosuppressive medications, including oral corticosteroids or chronic use of inhaled high potency corticosteroids within 60 days prior to the screening visit, or alkylating agents, antimetabolites, or radiation therapy;
8. has received any blood products or immunoglobulin within 3 months prior to first study injection, or is likely to require infusion of blood products during the study period;
9. has received the following prior to the first injection:
 - a. **≤28 days:**
 - any bacterial or viral vaccine
 - granulocyte or granulocyte-macrophage colony-stimulating factor, or
 - any other investigational medicinal agent
 - b. **at any time:** an injection of deoxyribonucleic acid plasmids or oligonucleotides;
10. is undergoing chemotherapy or expected to receive chemotherapy during the study period; has a diagnosis of cancer within the last 5 years other than squamous or basal cell carcinoma of the skin;
11. has a random non-fasting blood glucose ≥200 mg/dl at screening;
12. has any condition that in the opinion of the Investigator would interfere with compliance or with interpretation of the study results (e.g., substance abuse or alcohol abuse).

Study Endpoints

- Patterns of cellular responses and gene activation induced by Engerix-B
- Correlation of patterns of cellular and gene activation with anti-HBs antibody levels at various time points after injection

Statistical Methods The per-protocol (PP) population for analysis of study endpoints will comprise all subjects who receive all study injections and complete Visit 11.

A key objective of this study is to determine immune parameters that may correlate with the ability of the subject to mount a seroprotective response, defined as anti-HBs level of ≥ 10 mIU/mL, to a standard 3-dose regimen of Engerix-B. Previous studies have consistently shown that the response to Engerix-B declines progressively in adults over 40 years of age. In 2 studies performed by Dynavax, the fraction of subjects 40 to 55 years old that produced anti-HBs levels greater than or equal to 10 mIU/mL was 73% and 75%, respectively and, in one study, subjects 55 to 70 had a seroprotection rate of 56%. The goal in this study will be to maximize the number of potential non-responders. We therefore expect a study of up to 60 subjects aged 50 to 70 years old to yield approximately 12 to 20 non-responders, which can be compared to 40 to 48 responders. Sufficient data are unavailable to determine whether any of the variables investigated in the studies of cellular or gene expression changes will yield statistically significant results with these numbers, although many analogous comparison studies using microarray analyses have yielded significant data with study groups of approximately this size.

1 LIST OF ABBREVIATIONS AND DEFINITIONS OF TERMS

Abbreviation	Definition
AE	adverse event
Anti-HBs	anti-HBsAg antibody
BMI	body-mass index
BP	Blood pressure
CFR	Code of Federal Regulations
CRF	case report form
CTL	cytotoxic T lymphocyte
DNA	deoxyribonucleic acid
GCP	Good Clinical Practice
HBsAg	hepatitis B surface antigen
HBcAg	hepatitis B core antigen
HBV	hepatitis B virus
HCV	hepatitis C virus
HHS	Health and Human Services
HIV	human immunodeficiency virus
HR	heart rate
ICF	informed consent form
ICH	International Conference on Harmonisation
Ig	immunoglobulin
IL-4	interleukin-4
IRB	Institutional Review Board
IU	international units
mcg	microgram
mg/dl	milligrams per deciliter
mRNA	messenger ribonucleic acid
mIU/mL	milli-International units per milliliter
mL	milliliter
NLRP	nucleotide-binding domain and leucine-rich-repeat-containing protein
PBMC	peripheral mononuclear blood cell
PDC	plasmacytoid dendritic cells
PP	per-protocol
rHBsAg	recombinant hepatitis B surface antigen
RNA	ribonucleic acid
RR	respiratory rate
TLR	Toll-like receptor
VAERS	Vaccine Adverse Event Reporting System

2 BACKGROUND AND RATIONALE

2.1 Background

Engerix-B, a licensed adult hepatitis B vaccine, consists of 20 micrograms (mcg) of recombinant hepatitis B surface antigen (rHBsAg) adsorbed to 0.50 mg of alum. The approved regimen for adults 20 years of age and older calls for 3 total injections, 1 injection each at 0, 1 and 6 months. Subjects with an antibody level of ≥ 10 mIU/mL are generally considered to be protected. Seroprotection rates are 10% to 20% after the first dose and approximately 60% to 70% after the second dose in healthy young adults^{1,2}. Mean antibody levels attained after 2 doses, however, are generally low (<200 mIU/mL). Over 95% of young adults develop protective levels after the third dose³, but the immunogenicity declines in adults over 40 years of age. In a Dynavax Phase 3 study in healthy adults, the seroprotection rate for Engerix-B after 3 doses was 78% among those aged 40 to 55 years, and 56% among those aged 56 to 70 years.

Seroprotection is mediated by antibody to the viral surface antigen, HBsAg, and there is a well-established correlation between antibody levels after immunization and protection from infection. For the studies described here, this means that correlations of antibody level with parameters such as gene expression or cellular activation may be directly correlated to functional protection from infection.

A number of conditions are associated with reduced response rates to the standard 3 dose regimen of HBV vaccination, resulting either in undetectable responses or very weak responses that do not reach 10 mIU/mL after the third dose. Among nominally healthy adults, age (>40 years), smoking, alcohol abuse and obesity are all correlated with reduced response rates⁴⁻⁶.

Virtually all arms of the immune system show alterations that progress with age, both in man and in laboratory animals⁷. These changes represent a complex mix of cell, intrinsic and environmental factors and result in increased risk of autoimmunity, increased susceptibility to infection and decreased responses to most vaccines. In general, memory responses are relatively well maintained and the impairments are particularly notable in the primary responses to neoantigens⁸. Age-related alterations in the function of B cells and "classical" T cell subsets have been extensively studied⁹⁻¹¹, but much less is known about the effects of age on innate immunity^{12,13} and more recently defined T cell subsets. Nearly all adult vaccines are less effective with increasing age¹⁴⁻¹⁶, often resulting in difficulty protecting older persons who are at increased risk of infection and severe disease. This age-related hypo-responsiveness is observed with live attenuated, and subunit vaccines, with or without alum adjuvant⁸. For HBV, the decline in vaccine efficacy is already apparent in 40-50 year olds and responsiveness declines progressively each decade thereafter¹⁷. Because so many aspects of immune function are altered with age, it has been difficult to identify changes that are most central to the impaired response to vaccines. Without a clearly defined target, development of vaccines and adjuvants to boost responses in the elderly has remained an empirical exercise.

Alum, a fine precipitate of hydrated aluminum salts, has long been thought to act primarily as a depot, producing slow, sustained release of adsorbed antigens and possibly as a means to enhance uptake into phagocytic cells¹⁸. Both actions may well contribute to the adjuvant activity of alum; however, they do not fully explain it. What was not recognized until very recently is the ability of alum to stimulate a key component of innate immunity, the nucleotide-binding domain and leucine-rich-repeat-containing protein (NLRP)-3 inflammasome^{19,20}. Thus, alum is now viewed more properly as an active immunostimulatory adjuvant, not just a passive formulation. However, much remains to be clarified about the mechanisms of immune stimulation by alum²¹ and the importance of inflammasome activation in the adjuvant activity of alum is controversial^{18,22}. In addition, all work to date has been done either in mouse models or with human cells *in vitro* and important mouse-human differences, such as the strong Th2 polarizing activity in mice but not humans, have not been explained.

Alum is the definitive “Th2 adjuvant” in mice, stimulating highly polarized Th2 T cell responses and Th2-associated immunoglobulin (Ig) isotypes (IgG1 and IgE) when used in conjunction with virtually any protein antigen. The pathway leading to the ultimately interleukin-4 (IL-4)-dependent Th2 responses triggered by alum remain to be fully elucidated. In man, alum-adjuvanted vaccines usually produce mixed Th1 and Th2 responses²³ and do not provoke significant IgE responses in most individuals.

The present study will address some of the outstanding questions about the immune response to Engerix-B among older adults. Blood samples will be collected from volunteers at multiple time points after vaccination and analyzed for patterns of gene expression; changes in populations of immune cells, including T and B lymphocytes and dendritic cells; and function of vaccine-specific T cells. These parameters will be correlated with the kinetics and magnitude of the antibody response with the intention of understanding the key factors that distinguish people who do or do not respond to the vaccine.

2.2 Rationale for Protocol

One of the most promising approaches for studying the function of the human immune system is to study responses to well-defined vaccines. Vaccines constitute virtually the only form of immune challenge that can be safely and ethically administered to humans for such studies. This study will be a comprehensive analysis at multiple time points of the response to the HBV vaccine, Engerix-B. The properties of this vaccine offer opportunities to address important questions about the regulation of human responses to defined antigens. Knowledge gained from this study will provide a much more complete picture of the cells and molecules that are required to make an antibody response in man and will, at the same time, inform one of the most challenging problems in vaccine development, the basis for reduced responsiveness associated with a number of health conditions, including increasing age.

3 STUDY OBJECTIVE

3.1 Primary Objectives

- Document the changes in cellular responses and gene expression patterns in serial blood samples collected from healthy older adults after vaccination with Engerix-B
- Determine which of these changes correlate with the development of protective levels of antibody (anti-HBsAg ≥ 10 mIU/mL) to HBV

3.2 Secondary Objective

- Compare and contrast the cellular responses and gene expression patterns in healthy older adults after the first vs. second injection with Engerix-B

4 INVESTIGATIONAL PLAN

4.1 Study Design

This is a single arm, open label study of up to 60 healthy adult subjects 50 to 70 years of age not previously vaccinated with a HBV vaccine. After providing informed consent and meeting eligibility criteria, subjects will receive a total of 3 injections of Engerix-B per the recommended adult schedule, administered at Weeks 0, 4, and 24 (Visits 1, 5 and 10). After each of the first 2 injections (Visit 1 and Visit 5), subjects will be asked to return at specified times for the collection of blood samples for the investigational studies. After the third injection, samples will only be collected for the measurement of antibody response to HBV.

Samples will be collected from subjects at key time points after each of the first 2 injections. The principal assays used will be a global analysis of changes in gene expression analyzed on whole genome microarrays and changes in lymphocyte and dendritic cell numbers and activation states by flow cytometry. In addition, the function of T cells will be assessed by restimulation with HBV antigen and measurement of cytokine production. Both techniques are well established and are expected to provide important insights into the human response to these vaccines. Additional exploratory assays will be performed on selected samples, including analysis of serum cytokine levels and responses to a panel of innate immune system activators. DNA samples will be collected for studies to explore the possible genetic basis of vaccine responsiveness. Changes observed on days 1 and 3 after injection will reflect primarily the response of the innate immune system to the adjuvant itself, whereas samples 7 and 28 days after injection will largely reflect activities of T and B lymphocytes.

4.1.1 Study Treatment

Engerix-B (20 mcg HBsAg, GlaxoSmithKline) is a vaccine indicated for protection against infection caused by all known subtypes of HBV.

4.1.2 Selection of Study Population

The population for this study is healthy HBV naïve males and females 50 to 70 years of age, inclusive.

4.1.3 Study Period

Enrollment and signing of informed consent forms will occur between 1 and 21 days prior to the first injection of vaccine. Each subject will be given the 3-dose vaccination regimen (0, 4, and 24 weeks) as recommended by the manufacturer. Blood samples will be collected immediately prior to each of the first 2 injections of Engerix-B and at 1, 3, 7, and 28 days after those injections. Each subject is expected to remain in the study through Week 28 (Visit 11), 4 weeks after the third Engerix-B injection.

4.1.4 Rationale for Selection of Dose

The dose and schedule of Engerix-B (HBsAg concentration 20 mcg/mL at Weeks 0, 4, and 24 [Visits 1, 5, and 10]) are based on the approved indication for healthy subjects greater than 19 years of age as defined in the manufacturer's Prescribing Information (Appendix 2).

4.2 Eligibility Criteria

4.2.1 Inclusion Criteria:

A subject must meet all of the following inclusion criteria to participate in the study:

1. be 50 to 70 years of age, inclusive;
2. be in good health in the opinion of the Investigator, based upon medical history, physical examination, and laboratory evaluation;
3. be able to comprehend and follow all required study procedures, and be available for all visits scheduled in the study;
4. be serum negative for hepatitis B surface antigen (HBsAg), HBsAg antibody (anti-HBs), antibody to hepatitis B core antigen (anti-HBcAg), hepatitis C virus (HCV), and human immunodeficiency virus (HIV);
5. if female of childbearing potential, agree to consistently use a highly effective method of birth control from the screening visit through Visit 11^a;
6. be able and willing to provide informed consent.

4.2.2 Exclusion Criteria:

A subject who meets any of the following criteria is not permitted to participate in the study:

^a Highly effective methods of birth control include: chemical contraceptive, intrauterine device (IUD), or condom plus spermicide. Abstinence will also be accepted but only with written confirmation from the subject.

1. if female of childbearing potential, is pregnant, breastfeeding, or planning a pregnancy;
2. has a known history of multiple sclerosis;
3. has previously received any hepatitis B vaccine (approved or investigational);
4. has a history of sensitivity to any component of Engerix-B;
5. has body-mass index (BMI) >30, diabetes mellitus (Type I or II), or other chronic medical condition that in the opinion of the Investigator might interfere with the immune response to hepatitis B vaccination;
6. is unwilling or unable to comply with all the requirements of the protocol;
7. has current or suspected immunodeficiency disorder or treatment with immunosuppressive medications, including oral corticosteroids or chronic use of inhaled high potency corticosteroids within 60 days prior to the screening visit, or alkylating agents, antimetabolites, or radiation therapy;
8. has received any blood products or immunoglobulin within 3 months prior to first study injection, or is likely to require infusion of blood products during the study period;
9. has received the following prior to the first injection:
 - a. **≤28 days:**
 - any bacterial or viral vaccine,
 - granulocyte or granulocyte-macrophage colony-stimulating factor, *or*
 - any other investigational medicinal agent
 - b. **at any time:** an injection of deoxyribonucleic acid plasmids or oligonucleotides;
10. is undergoing chemotherapy or expected to receive chemotherapy during the study period; has a diagnosis of cancer within the last 5 years, other than squamous or basal cell carcinoma of the skin;
11. has a random non-fasting blood glucose ≥ 200 mg/dl at screening;
12. has any condition that in the opinion of the Investigator would interfere with compliance or with interpretation of the study results (e.g., substance abuse or alcohol abuse).

4.2.3 Removal of Subjects from Assessment

Subjects may be removed from the study for a variety of reasons, including the following:

1. Noncompliance with study procedures as determined by the Investigator or Dynavax,
2. Dynavax's administrative decision to terminate the study.

The primary reason for early discontinuation will be recorded on the appropriate case report form (CRF).

If a subject discontinues study injections early (prior to Visit 10), no further follow-up assessments or blood sample collection are required.

Subjects may choose to withdraw from the study at any time. The Investigator should discuss with Dynavax in all cases of potential subject withdrawal, whether initiated by the Investigator or subject, if it is appropriate for the Investigator to offer subsequent doses of Engerix-B in order to complete the standard vaccine series.

4.3 Treatment of Subjects

4.3.1 Treatment Administered

Subjects will be administered Engerix-B, 1.0 mL (20 mcg/mL) per the manufacturer's Prescribing Information (see Appendix 2).

4.3.2 Storage and Handling Instructions

Engerix-B will be stored, prepared, and dosed according to the manufacturer's Prescribing Information (see Appendix 2).

4.3.3 Directions for Administration of Engerix-B

Subjects will receive 1.0 mL Engerix-B (20 mcg HBsAg) administered intramuscularly in the deltoid muscle on Week 0 (Visit 1), Week 4 (Visit 5), and Week 24 (Visit 10). Each injection should be given in the alternate arm from the previous injection.

4.3.4 Blinding

This is an open-label study of the immune response to Engerix-B, hence, there is no blinding at the site for administration of Engerix-B. All specimens will be submitted to the laboratory with a subject identification number. The laboratory conducting the immunologic testing will be blinded to the subject's anti-HBs test results.

Subjects may be provided the result of their anti-HBs levels after they complete Visit 11.

4.3.5 Prohibited Treatments or Therapies

Prohibited medications include the following:

- **28 days prior to Injection 1 (Visit 1) and through Week 28 (Visit 11):**
 - any bacterial or viral vaccine
 - granulocyte or granulocyte-macrophage colony-stimulating factor
 - any other investigational medicinal agent
- **60 days prior to Screening and through Week 28 (Visit 11):** systemic corticosteroids (more than 3 consecutive days) or other immunomodulators or immune suppressive medication (such as chemotherapy), with the exception of inhaled steroids

- **90 days prior to Injection 1 (Visit 1) and through Week 28 (Visit 11):**
immunoglobulin
- **at any time prior to Injection 1 (Visit 1) and through Week 28 (Visit 11):**
injection of DNA plasmids or oligonucleotides

If a subject requires treatment that is prohibited, the Investigator should contact Dynavax to determine whether the subject should receive further study injections. If study injections are discontinued prior to Visit 10, the subject will be withdrawn from the study.

5 MANAGEMENT OF STUDY TREATMENT TOXICITIES

5.1 Stopping Rules

The protocol may be discontinued by Dynavax at any time (Section 8, Study Termination).

5.2 Management of Subject Safety

Post vaccination symptoms will be treated as appropriate by the Investigator. Adverse events assessed by the Investigator as related to administration of Engerix-B or other study procedures should be reported. See Section 11 (Investigator's Responsibilities) for additional details.

Injection-site reactions are expected to spontaneously subside. Local pruritus and pain can be treated with oral medications. If significant symptoms of pain and induration persist for more than 12 hours, an ice pack may be applied locally for 30 minutes every 2 hours, as needed. Use of an ice pack prior to 12 hours after the onset of symptoms is discouraged, as it may interfere with the action of the study vaccine. Avoid use of nonsteroidal anti-inflammatory drugs for 24 hours after study injection. Do not inject into a site if local pain, tenderness, swelling, or pruritus persists from a previous injection or other cause.

Epinephrine and other appropriate agents used for the control of immediate allergic reactions must be immediately available should an acute anaphylactic reaction occur.

To avoid diagnostic confusion between manifestations of an acute illness and possible vaccine adverse effects, vaccination with Engerix-B should be postponed in subjects with moderate or severe acute illness unless they are at immediate risk of hepatitis B infection.

6 STUDY PROCEDURES

6.1 Overview

The following sections outline the procedures to be performed for each study visit. Subjects should be encouraged to complete the entire study. All evaluations and blood specimens required at specified time points are listed in Appendix 1.

Blood will be collected for multiple assessments of humoral and cellular immune response using the following parameters. Timings for blood collection after each study

injection are based on results of preliminary analyses of the immune responses to vaccines.

- Serum will be collected at multiple times for:
 - o Measurement of anti-HBs levels at baseline, prior to the second and third injections, and at the last subject visit;
 - o A reserve serum aliquot will be reserved for potential analysis of serum cytokine levels
- Whole blood will be collected for genome-wide transcriptional profiling of mRNA.
- A separate blood sample will be collected for analysis of genetic polymorphisms.
- PBMCs will be collected for:
 - o Examination of the transcriptional response to TLR ligands;
 - o Phenotyping of B and T cells and dendritic cells;
 - o Antigen restimulation and phenotyping of antigen-specific T cells.

Starting at Day 1/ Visit 2, all subjects should complete and sign the patient questionnaire at each visit prior to other study procedures. Authorized study site personnel should review the information reported by the subject at each visit (prior to administration of study injection at Visits 5 and 10), including any changes in concomitant medications, to ensure no prohibited medications have been taken by the subject since the previous visit (Section 4.3.5, Prohibited Treatments or Therapies).

6.2 Duration of Study

Each randomized subject is expected to remain in the study through Week 28. Subjects will receive Engerix-B on Days 0, 28, and 168 (Weeks 0, 4, and 24). After completing Day 0/ Visit 1, all subjects will return to the site at Days 1, 3, 7, 28, 29, 31, 35, 56, 168, and 196 to have blood drawn for laboratory studies.

6.3 Visit Windows

The window for study injections is based off the first injection administered at Day 0/ Visit 1. Day 28/ Visit 5 may occur between Days 28 and 35, i.e., the visit window is +7 days from the target date after Day 0. Day 56/ Visit 9, Day 168/ Visit 10, and Day 196/ Visit 11 each have visit windows ± 7 days from the target date. All other visits must occur at the exact intervals as defined after the previous injection, i.e., Visit 7 must occur exactly 3 days after Visit 5.

6.4 Subject Information and Screening

All subjects must provide written informed consent before undergoing any study-related procedures. All subjects who sign the informed consent form (ICF) should be recorded in the screening log.

6.5 Day -21 to Day 0 / Screening Visit

The screening evaluations must be performed within 21 days (3 weeks) prior to the Day 0 visit.

- Review the ICF with each prospective subject to be certain the subject understands the purpose, design, methods, procedures, requirements, and risks of the study. Subjects who wish to participate in the study will be asked to sign the ICF and must do so before any protocol-specific screening procedures are performed.
- Review inclusion and exclusion criteria and confirm that the subject is eligible to participate.
- Review the subject's demographic information, medical and surgical history, and medication history (including vaccinations and current medications).
- Perform a brief physical examination targeted by subject reported symptoms, including height and weight assessment, check of general appearance, and systems evaluation. The brief physical examination should be performed by the Investigator or designee qualified to conduct physical examinations.
- Obtain vital signs (heart rate [HR], blood pressure [BP], respiratory rate [RR], oral temperature).
- Perform urine pregnancy test for all females of child-bearing potential. Urine dipstick can be used.
- Obtain blood samples:
 - o Random non-fasting blood glucose screening test
 - o Serum for hepatitis (includes HBsAg, anti-HBs, antibodies to HBcAg, HCV) and HIV screen
 - o Serum for laboratory assessments (follow instructions in Laboratory Manual for sample collection, storage, processing, and shipment):
 - Serum for measurement of anti-HBs antibody
 - Reserve serum aliquot

6.6 Day 0 / Visit 1: Administration of injection 1 (Engerix-B) and collection of blood samples

Visit 1 is the day of first injection of Engerix-B. The following procedures will be performed in the order noted below:

- Review inclusion and exclusion criteria and confirm that the subject is eligible to participate.
- Review interim medical and medication history (including vaccinations and current medications) to determine if the subject can receive the injection.
- Perform urine pregnancy test for all females of child-bearing potential. The urine pregnancy test must be negative in order for the subject to receive an injection. Urine dipstick can be used.
- Obtain blood samples prior to injection:
 - o Reserve serum aliquot
 - o Whole blood for RNA assays
 - o PBMCs

Note: Follow instructions in the Laboratory Manual for collection, storage, processing, and shipment.

- Inject Engerix-B according to the manufacturer's Prescribing Information.

6.6 Day 1 / Visit 2 (must occur 1 day after first injection): Collection of blood samples

- Administer patient questionnaire and review information reported by the subject.
- Obtain blood samples:
 - o Reserve serum aliquot
 - o Whole blood for RNA assays
 - o PBMCs

Note: Follow instructions in the Laboratory Manual for collection, storage, processing, and shipment.

6.7 Day 3 / Visit 3 (must occur 3 days after first injection): Collection of blood samples

- Administer patient questionnaire and review information reported by the subject.
- Obtain blood samples:
 - o Whole blood for RNA assays
 - o PBMCs

Note: Follow instructions in the Laboratory Manual for collection, storage, processing, and shipment.

6.8 Day 7 / Visit 4 (must occur 7 days after first injection): Collection of blood samples

- Administer patient questionnaire and review information reported by the subject.
- Obtain blood samples:

- o Whole blood for RNA assays
- o PBMCs

Note: Follow instructions in the Laboratory Manual for collection, storage, processing, and shipment.

6.9 Day 28 / Visit 5: Administration of injection 2 (Engerix-B) and collection of blood samples

Visit 5 is the day of second injection of Engerix-B. The following procedures will be performed in the order noted below:

- Administer patient questionnaire and review information reported by the subject, including any reported changes to interim medical history or medications (including vaccinations) to determine if the subject can receive the injection.
- Perform urine pregnancy test for all females. The urine pregnancy test must be negative in order for the subject to receive an injection. Urine dipstick can be used.
- Obtain blood samples prior to injection:
 - o Serum for measurement of anti-HBs
 - o Reserve serum aliquot
 - o Whole blood for RNA assays
 - o PBMCs

Note: Follow instructions in the Laboratory Manual for collection, storage, processing, and shipment.

- Inject Engerix-B according to the manufacturer's Prescribing Information.

6.10 Day 29 / Visit 6 (must occur 1 day after second injection): Collection of blood samples

- Administer patient questionnaire and review information reported by the subject.
- Obtain blood samples:
 - o Whole blood for RNA assays
 - o PBMCs

Note: Follow instructions in the Laboratory Manual for collection, storage, processing, and shipment.

6.11 Day 31 / Visit 7 (must occur 3 days after second injection): Collection of blood samples

- Administer patient questionnaire and review information reported by the subject.
- Obtain blood samples:
 - o Whole blood for RNA assays

- o Blood for genetic testing
- o PBMCs

Note: Follow instructions in the Laboratory Manual for collection, storage, processing, and shipment.

6.12 Day 35 / Visit 8 (must occur 7 days after second injection): Collection of blood samples

- Administer patient questionnaire and review information reported by the subject.
- Obtain blood samples:
 - o Whole blood for RNA assays
 - o PBMCs

Note: Follow instructions in the Laboratory Manual for collection, storage, processing, and shipment.

6.13 Day 56 / Visit 9: Collection of blood samples

- Administer patient questionnaire and review information reported by the subject.
- Obtain the following blood samples:
 - o Serum for measurement of anti-HBs
 - o Reserve serum aliquot
 - o Whole blood for RNA assays
 - o PBMCs

Note: Follow instructions in the Laboratory Manual for collection, storage, processing, and shipment.

6.14 Day 168 / Visit 10: Administration of injection 3 (Engerix-B) and collection of blood samples

Visit 10 is the day of third injection of Engerix-B. The following procedures will be performed in the order noted below:

- Administer patient questionnaire and review information reported by the subject, including any reported changes to interim medical history or medications (including vaccinations) to determine if the subject can receive the injection.
- Perform urine pregnancy test for all females. The urine pregnancy test must be negative in order for the subject to receive an injection. Urine dipstick can be used.
- Obtain the following blood samples prior to injection:
 - o Serum for measurement of anti-HBs
 - o Reserve serum aliquot

Note: Follow instructions in the Laboratory Manual for collection, storage, processing, and shipment.

- Inject Engerix-B according to the manufacturer's Prescribing Information, (Appendix 2).

6.15 Day 196 / Visit 11: Collection of blood samples

- Administer patient questionnaire and review information reported by the subject
- Obtain the following blood samples:
 - o Serum for measurement of anti-HBs antibody
 - o Reserve serum aliquot

Note: Follow instructions in the Laboratory Manual for collection, storage, processing, and shipment.

7 IMMUNOLOGIC VARIABLES

7.1 Study Endpoints

- Patterns of cellular responses and gene activation induced by Engerix-B
- Correlation of patterns of cellular and gene activation with anti-HBsAg antibody levels at various time points after injection

7.2 Appropriateness of Measurements

The measures of immune response employed here are experimental; however, most are well established techniques that are expected to produce meaningful and interpretable results. Specifically, monitoring of gene expression patterns with microarrays and measuring T cell responses to defined antigens are well established and frequently used as exploratory measures in clinical studies. Serum cytokine and innate response measures are less frequently used in clinical settings but common experimental techniques. In aggregate, these techniques are expected to provide a comprehensive view of the principal changes in immune functions that accompany injection of a vaccine

8 STUDY TERMINATION

The end of study date is defined as the date of the last visit of the last participant. However, the Sponsor reserves the right to terminate the study at any time. If the study is terminated prior to the last study injection, all eligible subjects will be given the opportunity to receive Engerix-B to complete the standard vaccine series.

The duration of the clinical study, as defined by start of enrollment to the final subject visit, is expected to last approximately 8 months; however, the analysis of samples will require several years of additional laboratory studies, and samples may be held up to 5 years after completion of the clinical study.

9 DATA QUALITY ASSURANCE

Prior to initiation of the study, representatives from Dynavax will review with the site personnel information about protocol requirements, and training on study procedures, e.g., CRF completion, required supplies, laboratory processing of peripheral blood mononuclear cells (PBMCs).

10 STATISTICAL METHODS

10.1 General

All statistical tests based on demographic, immunogenicity, or safety data will be performed at the 5% level of significance.

No imputations will be made for missing data. Non-evaluable subjects will not be replaced.

10.2 Sample Size Considerations

A key objective of this study is to determine immune parameters that may correlate with the ability of the subject to mount a seroprotective response, defined as anti-HBs level of ≥ 10 mIU/mL, to a standard 3-dose regimen of Engerix-B. Previous studies have consistently shown that the response to Engerix-B declines progressively in adults over 40 years of age. In 2 studies performed by Dynavax, the fraction of subjects 40 to 55 years old that produced anti-HBs levels greater than or equal to 10 mIU/mL was 73% and 75%, respectively and, in 1 study, subjects 56 to 70 had a seroprotection rate of 56%. The goal in this study will be to maximize the number of potential non-responders. We therefore expect a study of up to 60 subjects aged 50 to 70 years old to yield between approximately 12 to 20 non-responders that can be compared to 40 to 48 responders. Sufficient data are unavailable to determine whether any of the variables investigated in the studies of cellular or gene expression changes will yield statistically significant results with these numbers, although many analogous comparison studies using microarray analyses have yielded significant data with study groups of approximately this size.

10.3 Blinding

This is an open-label study of the immune response to Engerix-B. All specimens will be submitted to the laboratory with a subject identification number. The laboratory conducting the immunologic testing will be blinded to the subject's anti-HBs level at all visits.

10.4 Study Analysis Population

The per-protocol (PP) population for analysis of study endpoints will comprise all subjects who receive all study injections and complete Visit 11.

If a subject is found after completion of the study to be ineligible, e.g., have received a prohibited medication, the details will be evaluated to determine whether it will be appropriate to keep them in the study analysis population.

11 INVESTIGATOR'S RESPONSIBILITIES

Investigators are responsible for monitoring the safety of subjects throughout the course of the study and for providing or referring subjects for appropriate medical care.

If the Investigator suspects an adverse reaction is related to Engerix-B administration, the Investigator should contact GlaxoSmithKline at 1-888-825-5249 or the Vaccine Adverse Event Reporting System (VAERS), at 1-800-822-7976 or <http://vaers.hhs.gov>. The event should also be reported to Dynavax via CRF.

If the Investigator suspects an adverse reaction is related to any other study procedures, the event should be reported to Dynavax via CRF.

12 ETHICS

12.1 Institutional Review Board

The protocol and informed consent documents must be reviewed and approved by an appropriately composed institutional review board (IRB). The study will not be initiated at a site until appropriate written IRB approval of the protocol, ICF, and all recruiting materials (if applicable) are obtained by the Investigator. Copies should be reviewed and approved by Dynavax prior to submission to the IRB. The Investigator will submit periodic reports on the progress of the study as required by the IRB, in accordance with applicable governmental regulations, and in agreement with the policy established by Dynavax. In addition, the Investigator will inform the IRB of any protocol amendments and administrative changes and will obtain appropriate written IRB approval of all protocol amendments.

12.2 Ethical Conduct of the Study

This study will be conducted in accordance with the protocol; CFR Title 21, Parts 11, 50, 54, 56, 312 Subpart D; the Declaration of Helsinki; GCP/ICH Guidance E6; and applicable local regulatory requirements.

12.3 Informed Consent

The Investigator is responsible for obtaining informed consent from each subject participating in the study in compliance with CFR Title 21, Part 50, and ICH guidelines. Prior to initiation of the study at the site, the ICF must be reviewed and approved by Dynavax and the governing IRB. The Investigator or authorized designee will discuss the purpose and pertinent details of the study with each subject, and the subject must understand, sign, and date the appropriate IRB approved ICF before undergoing any study-specific procedures. The ICF must be signed and personally dated by the subject and by the person who conducted the informed consent discussion. Additional signature requirements may exist. The original signed and dated ICF will be retained with the subject's study records, and a copy of the signed ICF will be given to the subject.

13 STUDY DOCUMENTATION AND RECORDKEEPING

13.1 Documentation of Data

13.1.1 Case Report Forms

Case report forms will be used to collect study information for enrolled subjects. When information is available, authorized study site personnel will carefully and accurately record the data on the CRF. The CRFs for each subject must be reviewed and approved by the Investigator.

13.1.2 Data Entry in Database

CRFs will be maintained in a study binder and sent to the Sponsor as directed.

13.2 Record Retention

The Investigator must retain all records relating to the conduct of this study (including subject's study records, signed consent forms, CRFs, all correspondence, and other supporting documentation) for one year after the end of this study.

The Investigator may withdraw from the responsibility of retaining records only after transferring custody of the records to another individual who will accept responsibility for them. A written notice of transfer must be provided to Dynavax prior to or no later than 10 days after transfer.

The Investigator must obtain written approval from Dynavax prior to the destruction of any records relating to the conduct of this study.

14 USE OF INFORMATION AND PUBLICATION

It is understood by the Investigator that the information generated in this study is the property of Dynavax. It is understood that the Investigator is obliged to provide Dynavax with complete test results, all study data, and access to all study records.

Results from the investigation shall not be made available to any third party by the Investigator or any of their staff.

15 REFERENCES

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Appendix 1 Schedule of Study Assessments

Visit	Screening	Visit 1	Visit 2	Visit 3	Visit 4	Visit 5	Visit 6	Visit 7	Visit 8	Visit 9	Visit 10	Visit 11
Week	- 3	0			1	4			5	8	24	28
Day (Visit Window)	- 21 to - 1	0	1	3	7	28 (+7)	29	31	35	56 (±7)	168 (±7)	196 (±7)
Informed Consent	X											
Inclusion/Exclusion Criteria	X	X										
Medical/Medication History (a)	X	X										
Brief Physical Exam (including Height, Weight)	X											
Vital Signs (BP, HR, RR, Oral Temp)	X											
Pregnancy Test (for all females of childbearing potential) (c)	X	X				X					X	
Random blood glucose screen	X											
Serum for HIV and Hepatitis Screen (b)	X											
Patient Questionnaire (a)			X	X	X	X	X	X	X	X	X	X
Anti-HBs Serum Aliquot (d)	X					X				X	X	X
Reserve Serum Aliquot (d)	X	X	X			X				X	X	X
Whole blood for RNA assays (d)		X	X	X	X	X	X	X	X	X		
Blood for genetic testing								X				
PBMC collection (d)		X	X	X	X	X	X	X	X	X		
Study injection with Engerix-B		X				X					X	

(a) Concomitant medications and significant interim medical history should be reviewed at screening and prior to administration of each study injection at Visits 1, 5 and 10.

(b) Samples to be sent to local laboratory to confirm status for HBsAg, anti-HBs, anti-HBcAg, HCV, HIV.

(c) Dipstick can be used. Urine pregnancy test must be performed and confirmed negative prior to study injection.

(d) To be collected prior to study injection at Visits 1, 5 and 10.

HIGHLIGHTS OF PRESCRIBING INFORMATION

These highlights do not include all the information needed to use ENGERIX-B safely and effectively. See full prescribing information for ENGERIX-B.

ENGRIX-B [Hepatitis B Vaccine (Recombinant)]

Suspension for Intramuscular Injection

Initial U.S. Approval: 1989

RECENT MAJOR CHANGES

Warnings and Precautions, Latex (5.1)	08/2010
Warnings and Precautions, Infants Weighing Less Than 2,000 g (5.2)	12/2010
Warnings and Precautions, Apnea in Premature Infants (5.3)	05/2010
Warnings and Precautions, Moderate or Severe Acute Illness (5.5)	12/2010

INDICATIONS AND USAGE

ENGRIX-B is a vaccine indicated for immunization against infection caused by all known subtypes of hepatitis B virus. (1)

DOSAGE AND ADMINISTRATION

- ENGRIX-B is administered by intramuscular injection. (2.1)
- Persons from birth through 19 years of age: A series of three doses (0.5 mL each) given on a 0-, 1-, 6-month schedule. (2.2)
- Persons 20 years of age and older: A series of three doses (1 mL each) given on a 0-, 1-, 6-month schedule. (2.2)
- Adults on hemodialysis: A series of four doses (2 mL each) given as a single 2 mL dose or as two 1 mL doses on a 0-, 1-, 2-, 6-month schedule. (2.2)

DOSAGE FORMS AND STRENGTHS

- ENGRIX-B is a sterile suspension available in the following presentations:
- 0.5 mL (10 mcg) single-dose vials and prefilled syringes (3)
- 1 mL (20 mcg) single-dose vials and prefilled syringes (3)

CONTRAINDICATIONS

Severe allergic reaction (e.g., anaphylaxis) after a previous dose of any hepatitis B-containing vaccine, or to any component of ENGERIX-B, including yeast. (4)

WARNINGS AND PRECAUTIONS

- ENGRIX-B is available in vials and 2 types of prefilled syringes. One type of prefilled syringe has a tip cap which may contain natural rubber latex. The other type has a tip cap and a rubber plunger which contain dry natural latex rubber. Use of these syringes may cause allergic reactions in latex sensitive individuals. (5.1, 16)
- Apnea following intramuscular vaccination has been observed in some infants born prematurely. Decisions about when to administer an intramuscular vaccine, including ENGERIX-B, to infants born prematurely should be based on consideration of the infant's medical status, and the potential benefits and possible risks of vaccination. (5.2)

ADVERSE REACTIONS

The most common solicited adverse events were injection-site soreness (22%) and fatigue (14%). (6.1)

To report SUSPECTED ADVERSE REACTIONS, contact GlaxoSmithKline at 1-888-825-5249 or VAERS at 1-800-822-7967 or www.vaers.hhs.gov.

DRUG INTERACTIONS

Do not mix ENGERIX-B with any other vaccine or product in the same syringe or vial. (7.1)

USE IN SPECIFIC POPULATIONS

- Safety and effectiveness of ENGERIX-B have not been established in pregnant women and nursing mothers. ENGERIX-B should only be given to a pregnant woman if clearly needed. (8.1, 8.3)
- Antibody responses are lower in persons older than 60 years of age than in younger adults. (8.5)

See 17 for PATIENT COUNSELING INFORMATION.

Revised: December 2010

FULL PRESCRIBING INFORMATION: CONTENTS*

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- Recommended Dose and Schedule
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15 REFERENCES

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*Sections or subsections omitted from the full prescribing information are not listed.

FULL PRESCRIBING INFORMATION

1 INDICATIONS AND USAGE

ENGRIX-B[®] is indicated for immunization against infection caused by all known subtypes of hepatitis B virus.

2 DOSAGE AND ADMINISTRATION

2.1 Preparation for Administration

Shake well before use. With thorough agitation, ENGRIX-B is a homogeneous, turbid white suspension. Do not administer if it appears otherwise. Parenteral drug products should be inspected visually for particulate matter and discoloration prior to administration, whenever solution and container permit. If either of these conditions exists, the vaccine should not be administered.

ENGRIX-B should be administered by intramuscular injection. The preferred administration site is the anterolateral aspect of the thigh for infants younger than 1 year and the deltoid muscle in older children (whose deltoid is large enough for an intramuscular injection) and adults.

ENGRIX-B may be administered subcutaneously to persons at risk of hemorrhage (e.g., hemophiliacs). However, hepatitis B vaccines administered subcutaneously are known to result in a lower antibody response. Additionally, when other aluminum-adsorbed vaccines have been administered subcutaneously, an increased incidence of local reactions including subcutaneous nodules has been observed. Therefore, subcutaneous administration should be used only in persons who are at risk of hemorrhage with intramuscular injections.

Do not inject intravenously or intradermally. ENGRIX-B should not be administered in the gluteal region; such injections may result in suboptimal response.

2.2 Recommended Dose and Schedule

Persons From Birth Through 19 Years of Age: Primary immunization for infants (born of hepatitis B surface antigen [HBsAg]-negative or HBsAg-positive mothers), children (birth through 10 years of age), and adolescents (11 through 19 years of age) consists of a series of three doses (0.5 mL each) given on a 0-, 1-, and 6-month schedule.

Persons 20 Years of Age and Older: Primary immunization for persons 20 years of age and older consists of a series of three doses (1 mL each) given on a 0-, 1-, and 6-month schedule.

Adults on Hemodialysis: Primary immunization consists of a series of four doses (2 mL each) given as a single 2 mL dose or two 1 mL doses on a 0-, 1-, 2-, and 6-month schedule. In hemodialysis patients, antibody response is lower than in healthy persons and protection may persist only as long as antibody levels remain above 10 mIU/mL. Therefore, the need for booster doses should be assessed by annual antibody testing. A 2 mL booster dose (as a single 2 mL dose

or two 1 mL doses) should be given when antibody levels decline below 10 mIU/mL.¹ [See *Clinical Studies (14.2).*]

Table 1. Recommended Dosage and Administration Schedules

Group	Dose^a	Schedules
Infants born of: HBsAg-negative mothers	0.5 mL	0, 1, 6 months
HBsAg-positive mothers ^b	0.5 mL	0, 1, 6 months
Children: Birth through 10 years of age	0.5 mL	0, 1, 6 months
Adolescents: 11 through 19 years of age	0.5 mL	0, 1, 6 months
Adults: 20 years of age and older	1 mL	0, 1, 6 months
Adults on hemodialysis	2 mL ^c	0, 1, 2, 6 months

Hepatitis B surface antigen = HBsAg

^a 0.5 mL (10 mcg); 1 mL (20 mcg).

^b Infants born to HBsAg-positive mothers should also receive hepatitis B immune globulin (HBIG) [see *Dosage and Administration (2.5)*].

^c Given as a single 2 mL dose or as two 1 mL doses.

2.3 Alternate Dosing Schedules

There are alternate dosing and administration schedules which may be used for specific populations (e.g., neonates born of hepatitis B–infected mothers, persons who have or might have been recently exposed to the virus, and travelers to high-risk areas) (Table 2). For some of these alternate schedules, an additional dose at 12 months is recommended for prolonged maintenance of protective titers.

Table 2. Alternate Dosage and Administration Schedules

Group	Dose^a	Schedules
Infants born of: HBsAg-positive mothers ^b	0.5 mL	0, 1, 2, 12 months
Children:		
Birth through 10 years of age	0.5 mL	0, 1, 2, 12 months
5 through 10 years of age	0.5 mL	0, 12, 24 months ^c
Adolescents:		
11 through 16 years of age	0.5 mL	0, 12, 24 months ^c
11 through 19 years of age	1 mL	0, 1, 6 months
11 through 19 years of age	1 mL	0, 1, 2, 12 months
Adults:		
20 years of age and older	1 mL	0, 1, 2, 12 months

Hepatitis B surface antigen = HBsAg

^a 0.5 mL (10 mcg); 1 mL (20 mcg).

^b Infants born to HBsAg-positive mothers should also receive hepatitis B immune globulin (HBIG) [see *Dosage and Administration (2.5)*].

^c For children and adolescents for whom an extended administration schedule is acceptable based on risk of exposure.

2.4 Booster Vaccinations

Whenever administration of a booster dose is appropriate, the dose of ENGERIX-B is 0.5 mL for children 10 years of age and younger and 1 mL for persons 11 years of age and older. Studies have demonstrated a substantial increase in antibody titers after booster vaccination with ENGERIX-B. See Section 2.2 for information on booster vaccination for adults on hemodialysis.

2.5 Known or Presumed Exposure to Hepatitis B Virus

Persons with known or presumed exposure to the hepatitis B virus (e.g., neonates born of infected mothers, persons who experienced percutaneous or permucosal exposure to the virus) should be given hepatitis B immune globulin (HBIG) in addition to ENGERIX-B in accordance with Advisory Committee on Immunization Practices recommendations and with the package insert for HBIG. ENGERIX-B can be given on either dosing schedule (0, 1, and 6 months or 0, 1, 2, and 12 months).

3 DOSAGE FORMS AND STRENGTHS

ENGRIX-B is a sterile suspension available in the following presentations:

- 0.5 mL (10 mcg) single-dose vials and prefilled TIP-LOK[®] syringes
- 1 mL (20 mcg) single-dose vials and prefilled TIP-LOK syringes

[See *Description (11)* and *How Supplied/Storage and Handling (16)*].

4 CONTRAINDICATIONS

Severe allergic reaction (e.g., anaphylaxis) after a previous dose of any hepatitis B-containing vaccine, or to any component of ENGERIX-B, including yeast, is a contraindication to administration of ENGERIX-B [see *Description (11) and How Supplied/Storage and Handling (16)*].

5 WARNINGS AND PRECAUTIONS

5.1 Latex

ENGRIX-B is available in vials and 2 types of prefilled syringes. One type of prefilled syringe has a tip cap which may contain natural rubber latex. The other type has a tip cap and a rubber plunger which contain dry natural latex rubber. Use of these syringes may cause allergic reactions in latex sensitive individuals. The vial stopper does not contain latex. [See *How Supplied/Storage and Handling (16)*.]

5.2 Infants Weighing Less Than 2,000 g

Hepatitis B vaccine should be deferred for infants weighing <2,000 g if the mother is documented to be HBsAg negative at the time of the infant's birth. Vaccination can commence at chronological age 1 month or hospital discharge. Infants weighing <2,000 g born to HBsAg-positive mothers or mothers of unknown HBsAg status should receive vaccine and hepatitis B immune globulin (HBIG) within 12 hours if HBsAg status cannot be determined; the birth dose should not be counted as the first dose in the vaccine series and it should be followed with a full 3 dose standard regimen (total of 4 doses).² [See *Dosage and Administration (2)*.]

5.3 Apnea in Premature Infants

Apnea following intramuscular vaccination has been observed in some infants born prematurely. Decisions about when to administer an intramuscular vaccine, including ENGERIX-B, to infants born prematurely should be based on consideration of the infant's medical status, and the potential benefits and possible risks of vaccination. For ENGERIX-B, this assessment should include consideration of the mother's hepatitis B antigen status and the high probability of maternal transmission of hepatitis B virus to infants born of mothers who are HBsAg positive if vaccination is delayed.

5.4 Preventing and Managing Allergic Vaccine Reactions

Prior to immunization, the healthcare provider should review the immunization history for possible vaccine sensitivity and previous vaccination-related adverse reactions to allow an assessment of benefits and risks. Epinephrine and other appropriate agents used for the control of immediate allergic reactions must be immediately available should an acute anaphylactic reaction occur. [See *Contraindications (4)*.]

5.5 Moderate or Severe Acute Illness

To avoid diagnostic confusion between manifestations of an acute illness and possible vaccine adverse effects, vaccination with ENGERIX-B should be postponed in persons with moderate or severe acute illness unless they are at immediate risk of hepatitis B infection (e.g., infants born of HBsAg-positive mothers).

5.6 Altered Immunocompetence

Immunocompromised persons may have a diminished immune response to ENGERIX-B, including individuals receiving immunosuppressant therapy.

5.7 Multiple Sclerosis

Although no causal relationship has been established, rare instances of exacerbation of multiple sclerosis have been reported following administration of hepatitis B vaccines. In persons with multiple sclerosis, the benefit of immunization for prevention of hepatitis B infection must be weighed against the risk of exacerbation of the disease.

5.8 Limitations of Vaccine Effectiveness

Hepatitis B has a long incubation period. ENGERIX-B may not prevent hepatitis B infection in individuals who had an unrecognized hepatitis B infection at the time of vaccine administration. Additionally, it may not prevent infection in individuals who do not achieve protective antibody titers.

6 ADVERSE REACTIONS

6.1 Clinical Trials Experience

Because clinical trials are conducted under widely varying conditions, adverse reaction rates observed in the clinical trials of a vaccine cannot be directly compared to rates in the clinical trials of another vaccine and may not reflect the rates observed in practice.

The most common solicited adverse events were injection site soreness (22%) and fatigue (14%).

In 36 clinical studies, a total of 13,495 doses of ENGERIX-B were administered to 5,071 healthy adults and children who were initially seronegative for hepatitis B markers, and healthy neonates. All subjects were monitored for 4 days post-administration. Frequency of adverse events tended to decrease with successive doses of ENGERIX-B.

Using a symptom checklist, the most frequently reported adverse events were injection site soreness (22%) and fatigue (14%). Other events are listed below. Parent or guardian completed forms for children and neonates. Neonatal checklist did not include headache, fatigue, or dizziness.

Incidence 1% to 10% of Injections: *Nervous System Disorders:* Dizziness, headache.

General Disorders and Administration Site Conditions: Fever ($>37.5^{\circ}\text{C}$), injection site erythema, injection site induration, injection site swelling.

Incidence $<1\%$ of Injections: *Infections and Infestations:* Upper respiratory tract illnesses.

Blood and Lymphatic System Disorders: Lymphadenopathy.

Metabolism and Nutrition Disorders: Anorexia.

Psychiatric Disorders: Agitation, insomnia.

Nervous System Disorders: Somnolence, tingling.

Vascular Disorders: Flushing, hypotension.

Gastrointestinal Disorders: Abdominal pain/cramps, constipation, diarrhea, nausea, vomiting.

Skin and Subcutaneous Tissue Disorders: Erythema, petechiae, pruritus, rash, sweating, urticaria.

Musculoskeletal and Connective Tissue Disorders: Arthralgia, back pain, myalgia, pain/stiffness in arm, shoulder, or neck.

General Disorders and Administration Site Conditions: Chills, influenza-like symptoms, injection site ecchymosis, injection site pain, injection site pruritus, irritability, malaise, weakness.

6.2 Postmarketing Experience

In addition to reports in clinical trials, worldwide voluntary reports of adverse events received for ENGERIX-B since market introduction (1990) are listed below. This list includes serious adverse events or events which have a suspected causal connection to components of ENGERIX-B.

The following adverse events have been identified during postapproval use of ENGERIX-B. Because these events are reported voluntarily from a population of unknown size, it is not always possible to reliably estimate their frequency or establish a causal relationship to the vaccine.

Infections and Infestations: Herpes zoster, meningitis.

Blood and Lymphatic System Disorders: Thrombocytopenia.

Immune System Disorders: Allergic reaction, anaphylactoid reaction, anaphylaxis. An apparent hypersensitivity syndrome (serum sickness-like) of delayed onset has been reported days to weeks after vaccination, including: arthralgia/arthritis (usually transient), fever, and dermatologic reactions such as urticaria, erythema multiforme, ecchymoses, and erythema nodosum.

Nervous System Disorders: Encephalitis, encephalopathy, migraine, multiple sclerosis, neuritis, neuropathy including hypoesthesia, paresthesia, Guillain-Barré syndrome and Bell's palsy, optic neuritis, paralysis, paresis, seizures, syncope, transverse myelitis.

Eye Disorders: Conjunctivitis, keratitis, visual disturbances.

Ear and Labyrinth Disorders: Earache, tinnitus, vertigo.

Cardiac Disorders: Palpitations, tachycardia.

Vascular Disorders: Vasculitis.

Respiratory, Thoracic and Mediastinal Disorders: Apnea, bronchospasm including asthma-like symptoms.

Gastrointestinal Disorders: Dyspepsia.

Skin and Subcutaneous Tissue Disorders: Alopecia, angioedema, eczema, erythema multiforme including Stevens-Johnson syndrome, erythema nodosum, lichen planus, purpura.

Musculoskeletal and Connective Tissue Disorders: Arthritis, muscular weakness.

General Disorders and Administration Site Conditions: Injection site reaction.

Investigations: Abnormal liver function tests.

7 DRUG INTERACTIONS

7.1 Concomitant Administration With Vaccines and Immune Globulin

ENGERIX-B may be administered concomitantly with immune globulin.

When concomitant administration of other vaccines or immune globulin is required, they should be given with different syringes and at different injection sites. Do not mix ENGERIX-B with any other vaccine or product in the same syringe or vial.

8 USE IN SPECIFIC POPULATIONS

8.1 Pregnancy

Pregnancy Category C

Animal reproduction studies have not been conducted with ENGERIX-B. It is also not known whether ENGERIX-B can cause fetal harm when administered to a pregnant woman or can affect reproduction capacity. ENGERIX-B should be given to a pregnant woman only if clearly needed.

8.3 Nursing Mothers

It is not known whether ENGERIX-B is excreted in human milk. Because many drugs are excreted in human milk, caution should be exercised when ENGERIX-B is administered to a nursing woman.

8.4 Pediatric Use

Safety and effectiveness of ENGERIX-B have been established in all pediatric age groups. Maternally transferred antibodies do not interfere with the active immune response to the vaccine. *[See Adverse Reactions (6) and Clinical Studies (14.1, 14.3, 14.4).]*

8.5 Geriatric Use

Clinical studies of ENGERIX-B used for licensure did not include sufficient numbers of subjects 65 years of age and older to determine whether they respond differently from younger subjects. However, in later studies it has been shown that a diminished antibody response and seroprotective levels can be expected in persons older than 60 years of age.³

11 DESCRIPTION

ENGERIX-B [Hepatitis B Vaccine (Recombinant)] is a sterile suspension of noninfectious hepatitis B virus surface antigen (HBsAg) for intramuscular administration. It contains purified surface antigen of the virus obtained by culturing genetically engineered *Saccharomyces cerevisiae* cells, which carry the surface antigen gene of the hepatitis B virus. The HBsAg expressed in the cells is purified by several physicochemical steps and formulated as a suspension of the antigen adsorbed on aluminum hydroxide. The procedures used to manufacture ENGERIX-B result in a product that contains no more than 5% yeast protein.

Each 0.5-mL pediatric/adolescent dose contains 10 mcg of HBsAg adsorbed on 0.25 mg aluminum as aluminum hydroxide.

Each 1-mL adult dose contains 20 mcg of HBsAg adsorbed on 0.5 mg aluminum as aluminum hydroxide.

ENGERIX-B contains the following excipients: Sodium chloride (9 mg/mL) and phosphate buffers (disodium phosphate dihydrate, 0.98 mg/mL; sodium dihydrogen phosphate dihydrate, 0.71 mg/mL).

ENGERIX-B is available in vials and 2 types of prefilled syringes. One type of prefilled syringe has a tip cap which may contain natural rubber latex. The other type has a tip cap and a rubber plunger which contain dry natural latex rubber. The vial stopper does not contain latex. *[See How Supplied/Storage and Handling (16).]*

ENGERIX-B is formulated without preservatives.

12 CLINICAL PHARMACOLOGY

12.1 Mechanism of Action

Infection with hepatitis B virus can have serious consequences including acute massive hepatic necrosis and chronic active hepatitis. Chronically infected persons are at increased risk for cirrhosis and hepatocellular carcinoma.

Antibody concentrations ≥ 10 mIU/mL against HBsAg are recognized as conferring protection against hepatitis B virus infection.¹ Seroconversion is defined as antibody titers ≥ 1 mIU/mL.

13 NONCLINICAL TOXICOLOGY

13.1 Carcinogenesis, Mutagenesis, Impairment of Fertility

ENGERIX-B has not been evaluated for carcinogenic or mutagenic potential, or for impairment of fertility.

14 CLINICAL STUDIES

14.1 Efficacy in Neonates

Protective efficacy with ENGERIX-B has been demonstrated in a clinical trial in neonates at high risk of hepatitis B infection.^{4,5} Fifty-eight neonates born of mothers who were both HBsAg-positive and hepatitis B “e” antigen (HBeAg)-positive were given ENGERIX-B (10 mcg/0.5 mL) at 0, 1, and 2 months, without concomitant hepatitis B immune globulin (HBIG). Two infants became chronic carriers in the 12-month follow-up period after initial inoculation. Assuming an expected carrier rate of 70%, the protective efficacy rate against the chronic carrier state during the first 12 months of life was 95%.

14.2 Efficacy and Immunogenicity in Specific Populations

Homosexual Men: ENGERIX-B (20 mcg/1 mL) given at 0, 1, and 6 months was evaluated in homosexual men 16 to 59 years of age. Four of 244 subjects became infected with hepatitis B during the period prior to completion of the 3-dose immunization schedule. No additional subjects became infected during the 18-month follow-up period after completion of the immunization course.

Adults with Chronic Hepatitis C: In a clinical trial of 67 adults 25 to 67 years of age with chronic hepatitis C, ENGERIX-B (20 mcg/1 mL) was given at 0, 1, and 6 months. Of the

subjects assessed at month 7 (N = 31), 100% responded with seroprotective titers. The geometric mean antibody titer (GMT) was 1,260 mIU/mL (95% Confidence Interval [CI]: 709, 2,237).

Adults on Hemodialysis: Hemodialysis patients given hepatitis B vaccines respond with lower titers, which remain at protective levels for shorter durations than in normal subjects. In a clinical trial of 56 adults who had been on hemodialysis for a mean period of 56 months, ENGERIX-B (40 mcg/2 mL given as two 1 mL doses) was given at 0, 1, 2, and 6 months. Two months after the fourth dose, 67% (29/43) of patients had seroprotective antibody levels (≥ 10 mIU/mL) and the GMT among seroconverters was 93 mIU/mL.

14.3 Immunogenicity in Neonates

In clinical studies, neonates were given ENGERIX-B (10 mcg/0.5 mL) at 0, 1, and 6 months or at 0, 1, and 2 months of age. The immune response to vaccination was evaluated in sera obtained one month after the third dose of ENGERIX-B.

Among infants administered ENGERIX-B at 0, 1, and 6 months, 100% of evaluable subjects (N = 52) seroconverted by month 7. The GMT was 713 mIU/mL. Of these, 97% had seroprotective levels (≥ 10 mIU/mL).

Among infants enrolled (N = 381) to receive ENGERIX-B at 0, 1, and 2 months of age, 96% had seroprotective levels (≥ 10 mIU/mL) by month 4. The GMT among seroconverters (N = 311) (antibody titer ≥ 1 mIU/mL) was 210 mIU/mL. A subset of these children received a fourth dose of ENGERIX-B at 12 months of age. One month following this dose, seroconverters (N = 126) had a GMT of 2,941 mIU/mL.

14.4 Immunogenicity in Children and Adults

Persons 6 Months Through 10 Years of Age: In clinical trials, children (N = 242) 6 months to 10 years of age were given ENGERIX-B (10 mcg/0.5 mL) at 0, 1, and 6 months. One to 2 months after the third dose, the seroprotection rate was 98% and the GMT of seroconverters was 4,023 mIU/mL.

Persons 5 Through 16 Years of Age: In a separate clinical trial including both children and adolescents 5 through 16 years of age, ENGERIX-B (10 mcg/0.5 mL) was administered at 0, 1, and 6 months (N = 181) or 0, 12, and 24 months (N = 161). Immediately before the third dose of vaccine, seroprotection was achieved in 92.3% of subjects vaccinated on the 0-, 1-, and 6-month schedule and 88.8% of subjects on the 0-, 12-, and 24-month schedule (GMT: 117.9 mIU/mL versus 162.1 mIU/mL, respectively, $P = 0.18$). One month following the third dose, seroprotection was achieved in 99.5% of children vaccinated on the 0-, 1-, and 6-month schedule compared to 98.1% of those on the 0-, 12-, and 24-month schedule. GMTs were higher ($P = 0.02$) for children receiving vaccine on the 0-, 1-, and 6-month schedule compared to those on the 0-, 12-, and 24-month schedule (5,687.4 mIU/mL versus 3,158.7 mIU/mL, respectively).

Persons 11 Through 19 Years of Age: In clinical trials with healthy adolescent subjects 11 through 19 years of age, ENGERIX-B (10 mcg/0.5 mL) given at 0, 1, and 6 months produced a seroprotection rate of 97% at month 8 (N = 119) with a GMT of 1,989 mIU/mL (N = 118, 95% CI: 1,318, 3,020). Immunization with ENGERIX-B (20 mcg/1 mL) at 0, 1, and

6 months produced a seroprotection rate of 99% at month 8 (N = 122) with a GMT of 7,672 mIU/mL (N = 122, 95% CI: 5,248, 10,965).

Persons 16 Through 65 Years of Age: Clinical trials in healthy adult and adolescent subjects (16 to 65 years of age) have shown that following a course of 3 doses of ENGERIX-B (20 mcg/1 mL) given at 0, 1, and 6 months, the seroprotection (antibody titers ≥ 10 mIU/mL) rate for all individuals was 79% at month 6 (5 months after second dose) and 96% at month 7 (1 month after third dose); the GMT for seroconverters was 2,204 mIU/mL at month 7 (N = 110).

An alternate 3-dose schedule (20 mcg/1 mL given at 0, 1, and 2 months) designed for certain populations (e.g., individuals who have or might have been recently exposed to the virus and travelers to high-risk areas) was also evaluated. At month 3 (1 month after third dose), 99% of all individuals were seroprotected and remained protected through month 12. On the alternate schedule, a fourth dose of ENGERIX-B (20 mcg/1 mL) at 12 months produced a GMT of 9,163 mIU/mL at month 13 (1 month after fourth dose) (N = 373).

Persons 40 Years of Age and Older: Among subjects 40 years of age and older given ENGERIX-B (20 mcg/1 mL) at 0, 1, and 6 months, the seroprotection rate 1 month after the third dose was 88% and the GMT for seroconverters was 610 mIU/mL (N = 50). In adults older than 40 years of age, ENGERIX-B produced anti-HBsAg antibody titers that were lower than those in younger adults.

14.5 Interchangeability With Other Hepatitis B Vaccines

A controlled study (N = 48) demonstrated that completion of a course of immunization with 1 dose of ENGERIX-B (20 mcg/1 mL) at month 6 following 2 doses of RECOMBIVAX HB[®] (10 mcg) at months 0 and 1 produced a similar GMT (4,077 mIU/mL) to immunization with 3 doses of RECOMBIVAX HB (10 mcg) at months 0, 1, and 6 (GMT: 2,654 mIU/mL). Thus, ENGERIX-B can be used to complete a vaccination course initiated with RECOMBIVAX HB.⁶

15 REFERENCES

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16 HOW SUPPLIED/STORAGE AND HANDLING

ENGRIX-B is available in single-dose vials and prefilled disposable TIP-LOK syringes (packaged without needles) (Preservative Free Formulation):

10 mcg/0.5 mL Pediatric/Adolescent Dose

NDC 58160-820-01 Vial (contains no latex) in Package of 10: NDC 58160-820-11

NDC 58160-820-43 Syringe (tip cap may contain latex) in Package of 10: NDC 58160-820-52

NDC 58160-820-32 Syringe (tip cap and plunger contain latex) in Package of 5: NDC 58160-820-46

NDC 58160-820-32 Syringe (tip cap and plunger contain latex) in Package of 10: NDC 58160-820-51

20 mcg/mL Adult Dose

NDC 58160-821-01 Vial (contains no latex) in Package of 10: NDC 58160-821-11

NDC 58160-821-43 Syringe (tip cap may contain latex) in Package of 5: NDC 58160-821-48

NDC 58160-821-43 Syringe (tip cap may contain latex) in Package of 10: NDC 58160-821-52

NDC 58160-821-32 Syringe (tip cap and plunger contain latex) in Package of 1: NDC 58160-821-32

NDC 58160-821-31 Syringe (tip cap and plunger contain latex) in Package of 5: NDC 58160-821-46

Store refrigerated between 2° and 8°C (36° and 46°F). Do not freeze; discard if product has been frozen. Do not dilute to administer.

17 PATIENT COUNSELING INFORMATION

- Inform vaccine recipients and parents or guardians of the potential benefits and risks of immunization with ENGERIX-B.
- Emphasize, when educating vaccine recipients and parents or guardians regarding potential side effects, that ENGERIX-B contains non-infectious purified HBsAg and cannot cause hepatitis B infection.
- Instruct vaccine recipients and parents or guardians to report any adverse events to their healthcare provider.
- Give vaccine recipients and parents or guardians the Vaccine Information Statements, which are required by the National Childhood Vaccine Injury Act of 1986 to be given prior to

immunization. These materials are available free of charge at the Centers for Disease Control and Prevention (CDC) website (www.cdc.gov/vaccines).

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